

CATHRYN (KATE) P. MURPHY

Ph.D. Candidate, Department of Chemistry, Northwestern University
Chicago, IL · (513) 291-1283 · CathrynMurphy2027@u.northwestern.edu
cathrynpmurphy.github.io · github.com/cathrynpmurphy

RESEARCH INTERESTS

Non-equilibrium statistical mechanics and stochastic thermodynamics; tensor network algorithms (MPS/DMRG, quantics, tensor cross interpolation) for high-dimensional stochastic master equations; rare events and large deviations; noise, dissipation, and reliability in mesoscopic electronic devices.

EDUCATION

Ph.D., Chemistry (Theoretical)

Expected Fall 2027

Northwestern University, Evanston, IL

Advisor: Prof. Todd Gingrich · Non-equilibrium statistical mechanics · GPA: 3.90

B.S., Biochemistry & Molecular Biology

May 2022

University of South Carolina, Columbia, SC

GPA: 3.98

RESEARCH EXPERIENCE

Graduate Researcher, Gingrich Group

Aug 2022 – Present

Department of Chemistry, Northwestern University

- Developed low-rank tensor network (MPS/DMRG) methods for solving high-dimensional stochastic master equations, targeting rare events and heavy-tailed distributions where Monte Carlo sampling is inefficient; benchmarked against Monte Carlo simulations.
- Independently proposed a new direction for the group: modeling thermal-noise bit-flip errors in CMOS circuits. Designed a basis expansion and Galerkin projection to make a $\sim 10^{14}$ -state problem tractable and quantified the dissipation–reliability tradeoff (first-author paper; open-source code).
- Designed a spin-off project modeling anti-ambipolar organic electrochemical transistors (OECTs), coupling noisy electrochemical reactions with charge transfer; tested candidate models against a collaborating group’s experimental data and showed that spatial resolution is required to reproduce the observed hysteresis.
- Developed a new procedure for computing committor functions with tensor networks, leveraging a physics-informed decomposition from the two slowest eigenvectors of the master equation and its adjoint (computed with DMRG and an implicit Euler numerical solver); used it to explain why rare-event rates scale differently with system size in one-dimensional reaction-diffusion and CMOS bits.
- Implemented gradient and parameter-sensitivity calculations within tensor networks; collaborated with Professor Christian Maes at Katholieke Universiteit (KU) Leuven on developing tensor network methods for the calculation of non-equilibrium heat capacities in the boundary-driven asymmetric simple exclusion process
- Formulated the chemical master equation for stochastic reaction–diffusion networks in MPS-compatible form via the Doi–Peliti formalism; showed that Doi–Peliti operators are consistent with statistical definitions of space–time cross-correlation functions, and built stochastic simulation algorithms to compute these correlations in systems with spatial and temporal patterning (basis of NSF GRFP award).
- Ran large-scale simulations (Monte Carlo, TDVP) and eigensolves (DMRG, implicit Euler) on HPC clusters; present results to collaborators weekly and at national conferences.

Undergraduate Researcher, Quantum Theory Group

2020 – 2022

University of South Carolina · Advisors: Profs. Sophya Garashchuk & Vitaly Rassolov

- Used DFT calculations to estimate transition-state energies and geometries for substituted organic aromatic molecular rotors.
- Developed a molecular-orbital mechanism for promoting rotor rotation based on a combination of steric and delocalization effects.

Undergraduate Researcher, Rural & Minority Health Research Center

2019 – 2021

University of South Carolina · Advisors: Profs. Whitney Zahnd & Jan Eberth

- Built a quantitative dataset from unstructured, inconsistently written state cancer-control policy documents; designed a coding scheme and analyzed compliance with national recommendations for addressing rural care disparities (2 publications).

PUBLICATIONS

- [1] **Murphy, C.**, Nicholson, S., Freitas, N., Penocchio, E., Gingrich, T. “Dissipation–Reliability Tradeoff for Stochastic CMOS Bits in Series.” *Under review, Physical Review E* (2026). arXiv:2603.04658.
- [2] **Murphy, C.**, Evans, S., Askelson, N., Eberth, J., Zahnd, W. E. “Extent of Inclusion of ‘Rural’ in Comprehensive Cancer Control Plans in the United States.” *Preventing Chronic Disease* **18**, 210091 (2021).
- [3] Zahnd, W. E., **Murphy, C.**, Knoll, M., Benavidez, G., Day, K. R., Ranganathan, R., Luke, P., Zgodic, A., Shi, K., Merrell, M. A., Crouch, E., Brandt, H. M., Eberth, J. “The Intersection of Rural Residence and Minority Race/Ethnicity in Cancer Disparities in the United States.” *Int. J. Environ. Res. Public Health* **18**(4), 1384 (2021).

TALKS

Invited talk, New Methods and Applications of Tensor Networks Workshop, Flatiron Institute, New York, NY Sept 2026
“How Does a Bit Flip? Transition Path Theory Meets Tensor Networks.”

APS March Meeting Mar 2026
“Dissipation Reliability Tradeoff for Stochastic CMOS Bits in Series”

POSTER PRESENTATIONS

“Reliability and Dissipation in Series Networks of Low-Voltage CMOS Bits”, Berkeley Statistical Mechanics Meeting, Berkeley, CA 2026

“Tensor Networks for Estimating Reliability in Stochastic Low-Powered CMOS Circuits,” Telluride School on Theoretical Chemistry, Telluride, CO 2025

“Tensor Networks for Estimating Reliability in Stochastic Low-Powered CMOS Circuits,” Diffusion Fundamentals XI, Evanston, IL 2025

“Unveiling Parameter-Dependent Complexity in Spatially Correlated Reaction–Diffusion Systems with Tensor Networks,” Catalyzing Gender Equity at Schrödinger, New York, NY 2024

“A Computational Analysis of Differentially Substituted Imide Rotors,” Southeastern Theoretical Chemistry Association Annual Conference 2022

HONORS AND AWARDS

NSF Graduate Research Fellowship 2024

Weinberg College Outstanding Graduate Student Teacher Award, Northwestern University 2024

Phi Lambda Upsilon Outstanding Teaching Assistant Award, Northwestern University 2023, 2024

American Chemical Society, South Carolina Section Outstanding Senior Award 2022

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas, Matplotlib), Julia (ITensors.jl, LinearAlgebra), MATLAB, Maple; Linux, SLURM/HPC clusters, Git
Methods	Probability and stochastic processes (Markov chains, Kolmogorov/master equations); tensor networks algorithms (DMRG, TDVP, cross interpolation, and quantized tensor trains); large deviations theory (CGFs, Laplace and Legendre transforms); numerical linear algebra (eigensolvers, matrix exponentials, low-rank tensor decompositions, basis transformations); Monte Carlo and rare-event estimation; sensitivity analysis and gradients; Doi–Peliti formalism; DFT
Languages	English (native), Spanish (proficient), Italian (learning)

TEACHING AND MENTORING

Teaching Assistant, Graduate Statistical Mechanics (*Northwestern University*) Fall 2026

Teaching Assistant, Advanced Inorganic Chemistry Aug 2022 – Dec 2023

Department of Chemistry, Northwestern University

- Prepared and administered assessments for a 200-student course on introductory atomic and molecular theory; delivered weekly review lectures to sections of 20–25 students and ran supplemental support sessions.
- Recognized with the Phi Lambda Upsilon Outstanding Teaching Assistant Award.

Graduate Mentor, Gingrich Group May – Aug 2023

Department of Chemistry, Northwestern University

- Mentored a visiting undergraduate from Cornell University in developing and analyzing stochastic simulation methods, including interpretation of Fourier-transformed data and space–time correlation functions.

SERVICE AND OUTREACH

Co-organizer, Theoretical Chemistry Seminar Series 2024 – Present

Northwestern University

Recruit and host visiting theoretical scientists.

Volunteer Tutor, Refugee Education 2023 – Present

SMiLE (2024–present); Girl Forward (2024–present); Foraging Opportunities for Refugees in America (2023–2024); Chicago, IL

- Tutor English literacy and mathematics for elementary-age children and AP Chemistry for high-school students in Chicago’s Afghan and Rohingya refugee communities, with attention to the effects of trauma on learning and the importance of trust.

NEAR Mentor 2023 – 2024

The Graduate School, Northwestern University

Advised prospective international applicants one-on-one on statements, deadlines, and application strategy.

Social Media Chair, Women in STEM and Engineering Research (WISER) 2022 – 2023

Northwestern University

Developed content promoting events and highlighting the work of women researchers in science and engineering.

REFERENCES

Prof. Todd Gingrich (Ph.D. advisor)

Department of Chemistry

Northwestern University

todd.gingrich@northwestern.edu